

**Supplementary Table 1: Demographics and allergic disease attributes of human study subject.**

| Patient ID | Gender | Age | Food allergies                     | Asthma       | Environmental Allergies | Eczema |
|------------|--------|-----|------------------------------------|--------------|-------------------------|--------|
| ID-1       | M      | 15  | no                                 | Intermittent | yes                     | yes    |
| ID-2       | F      | 6   | no                                 | Intermittent | no                      | yes    |
| ID-3       | M      | 9   | no                                 | Intermittent | yes                     | no     |
| ID-4       | M      | 13  | no                                 | Intermittent | yes                     | yes    |
| ID-5       | F      | 9   | peanut, tree nuts, lentil, kiwi    | Intermittent | yes                     | yes    |
| ID-6       | M      | 9   | milk, egg, peanut, tree nuts       | Intermittent | yes                     | yes    |
| ID-7       | M      | 7   | no                                 | Intermittent | no                      | no     |
| ID-8       | F      | 19  | tree nuts, shellfish               | Intermittent | yes                     | no     |
| ID-9       | M      | 11  | no                                 | Intermittent | yes                     | yes    |
| ID-10      | F      | 10  | no                                 | Intermittent | no                      | yes    |
| ID-11      | F      | 8   | milk, egg, peanut, tree nuts, fish | Intermittent | no                      | yes    |
| ID-12      | F      | 10  | no                                 | Intermittent | yes                     | yes    |
| ID-13      | F      | 3   | cashew, pistachio, milk            | Intermittent | yes                     | yes    |
| ID-14      | M      | 11  | no                                 | Intermittent | no                      | no     |
| ID-15      | M      | 7   | peanut, tree nuts                  | Intermittent | yes                     | no     |
| ID-16      | M      | 6   | no                                 | Intermittent | cat                     | no     |
| ID-17      | M      | 9   | no                                 | Intermittent | no                      | yes    |
| ID-18      | M      | 13  | no                                 | Intermittent | yes                     | yes    |
| ID-19      | M      | 2   | no                                 | Intermittent | no                      | yes    |
| ID-20      | F      | 12  | peanut, tree nuts                  | Intermittent | yes                     | no     |
| ID-21      | F      | 10  | no                                 | Intermittent | yes                     | no     |
| ID-22      | F      | 9   | scallop                            | Intermittent | yes                     | yes    |
| ID-23      | F      | 9   | no                                 | Intermittent | yes                     | no     |
| ID-24      | F      | 13  | no                                 | Intermittent | no                      | no     |
| ID-25      | M      | 7   | no                                 | Intermittent | yes                     | yes    |

|       |   |    |                                         |                 |      |     |
|-------|---|----|-----------------------------------------|-----------------|------|-----|
| ID-26 | F | 4  | milk, egg, peanut, tree nuts,<br>sesame | Intermittent    | yes  | yes |
| ID-27 | M | 11 | no                                      | Intermittent    | no   | no  |
| ID-28 | F | 9  | peach, carrot (OAS)                     | Intermittent    | yes  | no  |
| ID-29 | M | 8  | no                                      | Intermittent    | yes  | yes |
| ID-30 | M | 8  | no                                      | Intermittent    | yes  | yes |
| ID-31 | M | 10 | no                                      | Intermittent    | no   | no  |
| ID-32 | M | 3  | peanut, tree nuts, sesame               | Intermittent    | yes  | yes |
| ID-33 | M | 9  | no                                      | Intermittent    | yes  | no  |
| ID-34 | M | 16 | egg, peanut, tree nuts, sesame          | Mild Persistent | yes  | yes |
| ID-35 | M | 11 | shellfish                               | Mild Persistent | yes  | no  |
| ID-36 | F | 10 | no                                      | Mild Persistent | no   | no  |
| ID-37 | M | 10 | no                                      | Mild Persistent | no   | no  |
| ID-38 | M | 12 | no                                      | Mild Persistent | no   | no  |
| ID-39 | M | 9  | no                                      | Mild Persistent | no   | yes |
| ID-40 | F | 17 | no                                      | Mild Persistent | yes  | no  |
| ID-41 | F | 10 | peanut                                  | Mild Persistent | yes  | no  |
| ID-42 | F | 11 | no                                      | Mild Persistent | no   | no  |
| ID-43 | M | 15 | no                                      | Mild Persistent | dust | no  |
| ID-44 | F | 8  | no                                      | Mild Persistent | yes  | no  |
| ID-45 | M | 12 | no                                      | Mild Persistent | yes  | no  |
| ID-46 | M | 9  | no                                      | Mild Persistent | no   | no  |
| ID-47 | M | 12 | no (OAS)                                | Mild Persistent | yes  | yes |
| ID-48 | M | 8  | no                                      | Mild Persistent | yes  | yes |
| ID-49 | M | 6  | no                                      | Mild Persistent | yes  | no  |
| ID-50 | M | 7  | no                                      | Mild Persistent | yes  | no  |
| ID-51 | F | 8  | no                                      | Mild Persistent | no   | no  |
| ID-52 | F | 9  | no                                      | Mild Persistent | yes  | no  |

|       |   |    |                                               |                     |     |     |
|-------|---|----|-----------------------------------------------|---------------------|-----|-----|
| ID-53 | F | 10 | milk, egg, peanut, sesame, fish,<br>shellfish | Mild Persistent     | yes | yes |
| ID-54 | M | 8  | no                                            | Mild Persistent     | yes | no  |
| ID-55 | F | 12 | no                                            | Mild Persistent     | yes | yes |
| ID-56 | F | 3  | egg                                           | Mild Persistent     | no  | yes |
| ID-57 | F | 1  | peanut, tree nuts                             | Mild Persistent     | no  | yes |
| ID-58 | M | 7  | no                                            | Mild Persistent     | yes | no  |
| ID-59 | M | 5  | egg, peanut, tree nuts                        | Mild Persistent     | yes | no  |
| ID-60 | F | 10 | no                                            | Mild Persistent     | no  | no  |
| ID-61 | F | 6  | no                                            | Mild Persistent     | yes | no  |
| ID-62 | F | 10 | no                                            | Mild Persistent     | yes | yes |
| ID-63 | F | 10 | no                                            | Mild Persistent     | yes | yes |
| ID-64 | M | 8  | no                                            | Mild Persistent     | no  | no  |
| ID-65 | F | 8  | no                                            | Mild Persistent     | yes | no  |
| ID-66 | M | 6  | no                                            | Mild Persistent     | yes | no  |
| ID-67 | F | 6  | no                                            | Mild Persistent     | no  | no  |
| ID-68 | M | 9  | no                                            | Mild Persistent     | yes | no  |
| ID-69 | M | 7  | yes chilli peppers                            | Mild Persistent     | no  | no  |
| ID-70 | M | 9  | no                                            | Mild Persistent     | no  | no  |
| ID-71 | M | 12 | no                                            | Mild Persistent     | no  | no  |
| ID-72 | M | 9  | no                                            | Mild Persistent     | yes | no  |
| ID-73 | M | 8  | no                                            | Moderate Persistent | yes | no  |
| ID-74 | F | 15 | Peanut, tree nuts                             | Moderate Persistent | yes | yes |
| ID-75 | M | 11 | no                                            | Moderate Persistent | yes | no  |
| ID-76 | M | 9  | no                                            | Moderate Persistent | yes | yes |
| ID-77 | M | 6  | no                                            | Moderate Persistent | yes | yes |
| ID-78 | F | 8  | no                                            | Moderate Persistent | yes | yes |
| ID-79 | M | 13 | no                                            | Moderate Persistent | yes | no  |

|        |   |    |                                                          |                     |     |     |
|--------|---|----|----------------------------------------------------------|---------------------|-----|-----|
| ID-80  | F | 10 | shellfish                                                | Moderate Persistent | yes | no  |
| ID-81  | M | 14 | no                                                       | Moderate Persistent | yes | yes |
| ID-82  | F | 9  | egg, peanut, tree nuts, sesame, shellfish, wheat, banana | Moderate Persistent | yes | yes |
| ID-83  | F | 12 | no                                                       | Moderate Persistent | yes | yes |
| ID-84  | M | 7  | no                                                       | Moderate Persistent | yes | no  |
| ID-85  | F | 10 | milk, peanut, tree nuts                                  | Moderate Persistent | yes | yes |
| ID-86  | F | 11 | no                                                       | Moderate Persistent | yes | yes |
| ID-87  | M | 5  | no                                                       | Moderate Persistent | yes | no  |
| ID-88  | M | 11 | no                                                       | Moderate Persistent | yes | no  |
| ID-89  | M | 4  | peanut, tree nuts                                        | Moderate Persistent | yes | yes |
| ID-90  | M | 8  | no                                                       | Moderate Persistent | no  | no  |
| ID-91  | F | 21 | multiple                                                 | Moderate Persistent | yes | yes |
| ID-92  | F | 17 | no                                                       | Moderate Persistent | yes | no  |
| ID-93  | M | 10 | no                                                       | Moderate Persistent | yes | no  |
| ID-94  | M | 4  | peanut                                                   | Moderate Persistent | yes | yes |
| ID-95  | M | 4  | peanut, tree nuts                                        | Moderate Persistent | yes | yes |
| ID-96  | M | 5  | peanut, tree nuts                                        | Severe Persistent   | yes | yes |
| ID-97  | M | 9  | no                                                       | Severe Persistent   | yes | yes |
| ID-98  | F | 3  | peanut, tree nuts, orange                                | Severe Persistent   | yes | yes |
| ID-99  | M | 2  | no                                                       | Severe Persistent   | yes | no  |
| ID-100 | M | 12 | no                                                       | Severe Persistent   | yes | yes |
| ID-101 | M | 11 | cow milk, peanut, tree nuts                              | Severe Persistent   | yes | yes |
| ID-102 | F |    | no                                                       | Severe Persistent   | no  | no  |
| ID-103 | M | 12 | shellfish, soy, beef                                     | Severe Persistent   | yes | yes |
| ID-104 | F | 11 | no                                                       | Severe Persistent   | no  | no  |
| ID-105 | F | 11 | no                                                       | Severe Persistent   | yes | no  |
| ID-106 | M | 6  | no                                                       | Severe Persistent   | yes | yes |

|        |   |    |                                        |    |     |     |
|--------|---|----|----------------------------------------|----|-----|-----|
| ID-107 | F | 10 | banana                                 | no | yes | no  |
| ID-108 | F | 1  | egg                                    | no | no  | no  |
| ID-109 | M | 2  | egg, peanut                            | no | no  | no  |
| ID-110 | M | 2  | egg, peanut                            | no | no  | no  |
| ID-111 | F | 2  | egg, peanut, tree nuts                 | no | no  | no  |
| ID-112 | M | 11 | peanut, tree nuts                      | no | no  | no  |
| ID-113 | M | 14 | peanut, tree nuts                      | no | yes | no  |
| ID-114 | M | 2  | peanut, tree nuts, sesame              | no | yes | no  |
| ID-115 | F | 20 | peanut, tree nuts, sesame,<br>shellfis | no | yes | no  |
| ID-116 | M | 5  | peanut, tree nuts. Sesme               | no | yes | no  |
| ID-117 | F | 12 | pineapple                              | no | no  | no  |
| ID-118 | M | 10 | sesame                                 | no | yes | no  |
| ID-119 | F | 10 | shellfish                              | no | no  | no  |
| ID-120 | F | 6  | tree nuts                              | no | yes | no  |
| ID-121 | M | 10 | tree nuts                              | no | yes | no  |
| ID-122 | M | 9  | tree nuts                              | no | yes | no  |
| ID-123 | M | 10 | tree nuts?                             | no | yes | no  |
| ID-124 | F | 9  | unknown                                | no | yes | no  |
| ID-125 | M | 1  | egg                                    | no | yes | yes |
| ID-126 | M | 3  | egg, peanut                            | no | yes | yes |
| ID-127 | F | 5  | egg, peanut, tree nuts                 | no | yes | yes |
| ID-128 | F | 10 | peanut, tree nuts                      | no | yes | yes |
| ID-129 | F | 2  | peanut, tree nuts                      | no | yes | yes |
| ID-130 | M | 8  | peanut, tree nuts, sesame              | no | yes | yes |
| ID-131 | F | 10 | sesame                                 | no | yes | yes |
| ID-132 | F | 8  | no                                     | no | yes | yes |
| ID-133 | M | 11 | no                                     | no | yes | yes |

|        |   |    |    |    |     |     |
|--------|---|----|----|----|-----|-----|
| ID-134 | M | 6  | no | no | yes | yes |
| ID-135 | F | 11 | no | no | yes | yes |
| ID-136 | M | 10 | no | no | no  | yes |
| ID-137 | M | 6  | no | no | no  | yes |
| ID-138 | F | 8  | no | no | yes | yes |
| ID-139 | F | 8  | no | no | no  | no  |
| ID-140 | M | 10 | no | no | no  | no  |
| ID-141 | F | 8  | no | no | no  | no  |
| ID-142 | M | 2  | no | no | no  | no  |
| ID-143 | M | 2  | no | no | no  | no  |
| ID-144 | M | 11 | no | no | no  | no  |
| ID-145 | M | 1  | no | no | no  | no  |
| ID-146 | M | 8  | no | no | no  | no  |
| ID-147 | M | 10 | no | no | no  | no  |
| ID-148 | M | 9  | no | no | no  | no  |
| ID-149 | F | 9  | no | no | no  | no  |
| ID-150 | F | 9  | no | no | no  | no  |
| ID-151 | M | 11 | no | no | no  | no  |
| ID-152 | F | 10 | no | no | no  | no  |
| ID-153 | M | 8  | no | no | no  | no  |
| ID-154 | F | 8  | no | no | no  | no  |
| ID-155 | F | 8  | no | no | no  | no  |
| ID-156 | M | 9  | no | no | no  | no  |
| ID-157 | F | 7  | no | no | no  | no  |
| ID-158 | M | 9  | no | no | no  | no  |
| ID-159 | M | 1  | no | no | no  | no  |
| ID-160 | M | 12 | no | no | no  | no  |
| ID-161 | F | 7  | no | no | no  | no  |

|        |   |    |    |    |    |    |
|--------|---|----|----|----|----|----|
| ID-162 | F | 9  | no | no | no | no |
| ID-163 | M | 1  | no | no | no | no |
| ID-164 | M | 9  | no | no | no | no |
| ID-165 | F | 12 | no | no | no | no |
| ID-166 | F | 16 | no | no | no | no |
| ID-167 | M | 8  | no | no | no | no |
| ID-168 | F | 7  | no | no | no | no |
| ID-169 | F | 8  | no | no | no | no |
| ID-170 | M | 10 | no | no | no | no |
| ID-171 | M | 10 | no | no | no | no |
| ID-172 | F | 9  | no | no | no | no |
| ID-173 | F | 8  | no | no | no | no |
| ID-174 | F | 8  | no | no | no | no |
| ID-175 | M | 1  | no | no | no | no |
| ID-176 | F | 10 | no | no | no | no |
| ID-177 | F | 12 | no | no | no | no |
| ID-178 | M | 7  | no | no | no | no |
| ID-179 | F | 7  | no | no | no | no |
| ID-180 | M | 7  | no | no | no | no |
| ID-181 | F | 10 | no | no | no | no |
| ID-182 | M | 8  | no | no | no | no |
| ID-183 | F | 7  | no | no | no | no |